Roll Control System for SCRT Rocket

Final Report

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DISCLAIMER

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ABSTRACT

The Student Competition Rocket Team (SCRT) Roll Control project design is centered around providing an active roll control system for the SCRT rocket through the use of trim-tabs. Uncontrolled roll can negatively impact vehicle stability, sensor accuracy, and overall flight performance. This system will increase the stability of the rocket post-burn, by dynamically adjusting trim tab deflection to counteract undesired roll. The current design of the system houses each of the four servos into the rocket fincan, with each servo spline aligned with a trim tab located at the trailing edge of each rocket fin. Batteries and a protoboard will be mounted on the inside of the fincan to provide power and control signals to the system. The servos will attach to each trim tab through a clamp mechanism to deflect the trim tabs throughout flight, allowing the redirection of airflows to counteract the rocket's roll. The final design centers on servo-actuated trim tabs mounted to each fin by custom clamp parts, enabling controllable aerodynamic roll control. All mechanical and electrical components are securely integrated within the rocket's fincan, making the most of geometrical constraints.

All of the components have been manufactured or obtained through third parties, and the assembly has been completed and evaluated against project requirements. The fins have been machined to the required dimensions and finished with sanding operations, the system architecture has also been refined through manufactured housing for mechanical and electrical components. The trim tabs meet the required actuation response time and correctly adjusts deflection based on fincan orientation.

ACKNOWLEDGEMENTS

[Use this page for acknowledging those who have substantially supported or assisted you, such as faculty and staff members, fellow students, sponsor mentors, sponsors, companies that donated time or supplies, etc.]

The SCRT Roll Control Capstone team would like to thank the individuals whose support contributed to the completion of this project.

We would like to first extend our gratitude to our Professor, Elliot Clement, for consistently providing guidance and feedback throughout the design and development process, which was instrumental to the shaping of our final product.

We would also like to express our appreciation to the SCRT club and our mentors, Austin Hays, Max Bartnik, and Ben Amos, for their continued support and technical input throughout the duration of the Capstone project. We would also like to thank Grace, for working with us to obtain the necessary materials for this project through the club and external suppliers. The expertise, guidance, and assistance from the SCRT club was essential in guiding the direction for this project and supporting the application of engineering principles to a real-world system with new challenges and constraints.

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1 BACKGROUND

1.1 Introduction

The purpose of this project is to design, analyze, and prototype an active roll-control system for a sub-scale solid-fuel rocket as part of the AIAA Oregon State Student Competition Rocket Team (SCRT). The current rocket platform does not include an active roll stabilization mechanism, which can lead to uncontrolled rotation during ascent. This lack of roll control negatively impacts aerodynamic efficiency, flight stability, and overall mission performance.

As identified in sponsor discussions, the roll-control system is being developed from the ground up, with no prior prototype or completed trade studies available. [1] This creates a unique engineering challenge that requires a full design process, including concept generation, trade studies, analysis, and prototyping. The primary objective of this project is to develop a compact, reliable, and effective system capable of actively controlling roll while meeting strict geometric, safety, and competition constraints.

This project is of significant interest to the sponsor, SCRT, because improved roll stability directly enhances rocket performance and increases the likelihood of successful competition flights. A functional roll-control system enables better trajectory control, reduces aerodynamic inefficiencies, and improves payload accuracy. Additionally, the system must integrate seamlessly with existing rocket components without introducing excessive mass, complexity, or drag.

Beyond the immediate application, this project addresses broader contemporary issues in aerospace engineering, including the need for efficient, lightweight, and reliable control systems in increasingly constrained environments. As modern aerospace systems demand higher levels of performance and precision, compact actuation and control solutions, such as servo-driven aerodynamic surfaces, are becoming more relevant. This project reflects those trends by focusing on a scalable, modular control approach that can be applied to future rocket designs.

Upon completion, this project will provide SCRT with a validated roll-control system design and a working prototype that can be further tested and refined. It will also establish a foundation for future teams to build upon, contributing to long-term improvements in rocket stability, performance, and design capability.

1.2 Project Scope

The OSU Student Competition Rocket Team is developing a high-performance launch vehicle to compete in collegiate rocketry events requiring precision trajectory control, stable flight, and accurate recovery. In these competitions, uncontrolled roll motion can degrade flight performance, bias sensor measurements, introduce aerodynamic inefficiencies, and jeopardize mission objectives such as altitude accuracy or payload deployment.

The purpose of this project is to research, design, build, and assess a working prototype of an aerodynamic roll control system for use on a competition-level sounding rocket, operated through Oregon State University's Student Competition Rocket Team (SCRT). The system will provide controlled roll stabilization and/or roll rate authority during powered ascent using aerodynamic control surfaces and associated hardware and software. A sub-scale prototype will be designed, operated, and tested on SCRT's sub-scale Sounding Rocket, with the intent of being implemented into the full-scale model in the near future. The design is meant to be easily integrated with existing rocket systems, both hardware and software, to allow the team a full range of flexibility for future systems and operations.

Over the course of this project, we will encounter design challenges that pertain to the compatibility of existing rocket systems, durability & reusability, and hardware/software integration. With the assistance of our advisors, the SCRT leadership team and Capstone advisors, we hope to benefit the Student Competition Rocket Team and broader AIAA community with development of a system that can be reused for generations of aerospace applications at OSU.

2 DESIGN PROCESS

After completing preliminary studies and research of roll control options, the team began working on the design of the trim tab roll control system. Beginning with the fincan model provided by the SCRT, the team created individual concepts for trim tab integration, while simultaneously researching electronic parts. Over the course of the Winter term, the team remodeled the original fincan, paying close attention to geometrical constraints and preserving aerodynamic properties and minimizing the impact of mass. This iterative design process led to the development of integrated housing for all components and a manufacturable system capable of achieving the intended trim tab functionality.

The design process for the roll control subsystem balanced stability requirements, time and physical constraints, and budget. An intricate and sponsor-involved downselection process was used to refine and validate concepts prior to application. Multiple iterations have been completed, with each revision addressing limitations and issues from previous concepts to improve upon the manufacturability, integration, and functionality of the system. Work was distributed among team members according to skillsets and interest, enabling comprehensive and concurrent design development of both mechanical and electrical components. Regular design reviews were conducted and team meetings ensured that subsystems and integration of components were properly aligned with the project.

3 DESIGN PROPOSAL – First Term

Before analyzing specific aerodynamic control systems, the Capstone team had to clearly outline the Customer Requirements (CR) and parameters outlined by SCRT and the International Rocket Engineering Competition (IREC).

Firstly, for the purposes of the Capstone project and SCRT requirements, the capstone team aimed to design an active control system over a passive control system. This allows the greatest range of control flexibility for SCRT and offers the largest performance upgrade from the current, passive system consisting of simple fins. This allows the team to investigate a wide range of active systems, ranging from canards and Trim Tabs, to thrust vectoring and cold gas thrusters.

Secondly, per IREC competition requirements, rocketry teams are limited to using active control systems for only when “the launch vehicle’s boost phase has ended (i.e., all propulsive stages have ceased producing thrust)”[2]. This limits the selection of control systems to those that can be effective after all the propellant in the rocket has burned off, and therefore only the systems that can be effective without the use of propellant/thrust. This rules out the possibility of utilizing thrust vectoring systems for the design solution.

Finally, the established constraints outlined within the project scope and CRs will be considered along with the team’s trade studies. This includes the space constraints of both Sub-Scale and Competition rockets, including weight, cost, control authority, and system integration. After these considerations, the team chose four main control concepts to consider; Canards, Trim Tabs, Rollerons, and Differential Fin Controls.

4 Design Solution

The final design solution for the SCRT Roll Control project is a servo actuated trim tab system integrated into the rocket fincan. The system uses four movable trim tabs, with one trim tab located at the trailing edge of each fin, to generate aerodynamic rolling moments during the coast phase of flight. This design was selected because it provides active roll-control authority while requiring minimal changes to the existing airframe geometry. The trim tab system also avoids interference with forward-mounted avionics and guidance components, which made it more compatible with the SCRT rocket than other concepts such as canards or full differential fin control.

The final design is made up of four major subsystems: the fin and trim tab assembly, the servo actuation mechanism, the internal component mounting system, and the electrical control system. Together, these subsystems create a compact roll-control prototype that can be installed in the sub-scale rocket and further developed for future full-scale SCRT use. The completed design does not represent a fully flight qualified system, but it provides SCRT with a working mechanical and electrical architecture that can be tested, improved, and validated by future teams.

4.1 Description of Solution

The trim tab roll-control system is designed around the existing fincan structure of the SCRT rocket. Each fin includes a small trim tab positioned along the trailing edge. During flight, the trim tabs deflect slightly away from their neutral position. This deflection changes the airflow over the fin and creates a small aerodynamic force. When commanded in the correct direction, the combined effect of the trim tabs creates a rolling moment that can oppose unwanted rocket roll.

The trim tabs are intended to operate after motor burnout, during the portion of flight where active aerodynamic control is permitted and where the rocket still has enough velocity for the trim tabs to create meaningful aerodynamic forces. The flight version of the system would use roll-rate data from onboard inertial sensors to command trim tab motion. In the completed prototype, the main focus was to prove that the trim tabs, servos, and internal mounting system could be physically integrated into the rocket geometry and actuated in a controlled and repeatable way.

The final mechanical design uses one servo per fin. Each servo is mounted internally so that the output spline is aligned with the trim tab actuation mechanism. The servo output is connected to the trim tab through a compact clamp and coupler system. This mechanism transfers servo rotation to the trim tab while keeping the external aerodynamic profile as clean as possible. The connection was designed to fit within the limited available space near the fin and fincan while still allowing the trim tab to rotate through its required range of motion.

The fin and trim tab assembly was constrained by the thin fin geometry and the need to avoid large external protrusions. Because the fins are relatively thin, the actuation mechanism had to remain compact and lightweight. A shaft-supported trim tab design was used so the trim tab could rotate while remaining aligned with the fin surface. The clamp components were designed to secure the trim tab to the shaft or coupler while minimizing looseness in the mechanism. This was important because excessive play in the linkage would reduce control accuracy and could allow the trim tab to flutter or move unpredictably under aerodynamic loading.

The internal mounting system was designed to hold the major electrical and mechanical components securely inside the rocket structure. The servos are positioned near the fins to reduce linkage complexity and preserve direct actuation of the tabs. The battery housing and protoboard housing are mounted inside the rocket body so that the electronics remain protected and organized during handling and testing. The housings include mounting features for screws and fasteners so that the components can be removed, inspected, or replaced without redesigning the full system.

The electrical system is centered around providing controlled power and command signals to each servo. The protoboard supports the electrical connections needed for servo control, while the battery system provides independent power for the actuation system. The intended control architecture uses filtered inertial measurement unit data to determine the rocket roll rate and command the trim tabs toward a stabilizing position. For the current prototype, the system was evaluated primarily through bench testing and integration testing rather than closed-loop flight control.

A key design requirement was that the system remain near neutral during inactive, startup, or fail-safe conditions. A neutral trim tab position is important because an unintended tab deflection could introduce roll rather than correct it. The final design therefore emphasizes repeatable alignment of the servo horn, coupler, clamp, and trim tab. During assembly, the servos are centered before the mechanical linkage is attached so that the trim tabs can be physically aligned with the neutral fin position. This provides a defined starting point for future calibration and control-loop tuning.

The main advantage of this design is that it adds active roll-control capability without requiring a complete redesign of the rocket body. The trim tabs are small, lightweight, and mounted directly to the existing fins. The servos and electronics are placed internally, which protects them from external airflow and reduces aerodynamic disturbance. The design is also modular because individual servos, trim tabs, clamps, and housings can be modified or replaced without changing the entire roll-control architecture.

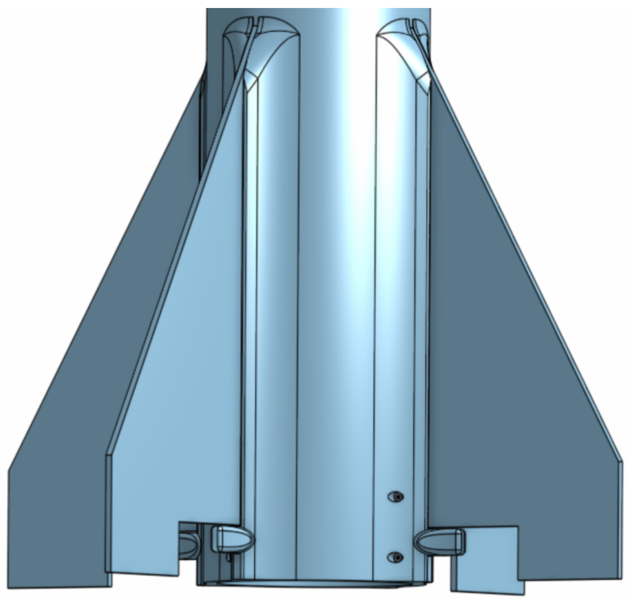

Figure 1. Final CAD model of the servo-actuated trim tab fincan assembly. Trim Tabs are transparent for positional reference.

4.2 Project Results

The final project result is a developed trim-tab roll-control prototype and supporting CAD package for SCRT. The team completed the major design work needed to define the final system layout, including the fincan assembly, trim tab placement, servo integration, internal housings, and actuation mechanism. The design was developed to satisfy the main project goal of creating an active aerodynamic roll-control system that can be further tested and refined by SCRT.

The team successfully down-selected to a trim-tab-based design after comparing the major feasible roll-control concepts. Canards offered strong control authority but created integration concerns with forward-mounted rocket systems. Rollerons provided passive roll damping but did not provide active command authority. Differential fin control offered strong authority but introduced greater mechanical complexity and higher fin-root loading concerns. Trim tabs provided the best balance of control authority, manufacturability, compactness, and compatibility with the existing rocket geometry. This allowed the team to move forward with a design that could realistically be prototyped within the time, budget, and manufacturing constraints of the Capstone project.

The CAD model showed that the trim tabs, servo mounts, internal electronics housings, and linkage components could fit within the available rocket geometry. This was one of the most important results of the project because the roll-control system had to fit inside a highly constrained rocket structure. The internal layout also showed that the servos could be positioned close enough to the trim tabs to avoid a long or complicated linkage system. This reduces the number of moving parts and improves the likelihood of reliable actuation.

The prototype manufacturing process focused on producing the custom parts needed to prove the mechanical fit and actuation concept. The team used 3D printed components for the early housings, clamps, and fit-check parts because this allowed rapid iteration and low-cost design changes. These parts were used to evaluate component clearances, mounting hole placement, trim tab movement, and assembly access. The use of 3D printed parts was especially useful because small changes to the clamp geometry, housing size, and mounting features could be made quickly after fit issues were identified.

During physical assembly, the team confirmed that the mechanism required careful alignment between the servo output, coupler, clamp, and trim tab. Any misalignment or looseness in this connection can create unwanted play in the trim tab. Early assembly work showed that thin wire or overly flexible linkage components allowed too much wobble in the trim tab connection. This was an important result because it showed that the final mechanism needs a stiffer and more repeatable connection method, such as a properly sized pin, shaft, or rigid coupler, rather than a flexible wire connection. This finding directly influenced the recommended future improvements for the actuation mechanism.

Bench-level testing showed that the servos could be commanded and used to rotate the trim tab mechanism through a small deflection range. This demonstrated that the basic actuation concept is feasible. However, the testing also showed that mechanical play, clamp stiffness, and linkage alignment are important factors that must be improved before the system can be considered flight-ready. The trim tabs do not need to move through a large angle to influence roll, but they do need to move consistently and return to neutral accurately. For this reason, repeatable neutral positioning and reduction of linkage wobble are two of the most important future verification items.

The team also evaluated the design against the major customer requirements and engineering specifications. The design meets the integration requirement because it fits within the rocket fincan layout and does not require major changes to the outer rocket body. It partially meets the roll-control authority requirement because analysis and design reasoning show that trim tabs can create controlled rolling moments, but the system has not yet been validated through wind tunnel or flight testing. It partially meets the response requirement because the selected servo-based architecture can provide active motion, but final response time and closed-loop control performance still require testing with the full avionics system. The system supports the safety requirement by emphasizing neutral trim tab alignment and internal mounting of components, but final safety validation must be completed before any launch.

Table 1 summarizes the current verification status of the major requirements.

Requirement / Objective	Verification Method	Result	Status
Provide active roll-control capability	Trade study, aerodynamic reasoning, trim tab design	Trim tabs selected as the best active aerodynamic solution for the SCRT rocket geometry	Met
Fit within rocket geometry	CAD integration and physical fit checks	Servos, trim tabs, housings, and internal components were arranged within the available fincan/body space	Met
Avoid interference with existing forward systems	Design review and layout comparison	Trim tabs are located on the fins and do not require forward-mounted control surfaces	Met
Actuate trim tabs from neutral	Servo bench testing and mechanical assembly	Servo-based actuation demonstrated	Met
Maintain neutral or safe position when inactive	Servo centering and trim tab alignment process	Neutral alignment approach established; final fail-safe behavior requires additional testing	Met
Support future SCRT development	CAD files, prototype parts, and design documentation	Final architecture can be handed off for future iteration and testing	Met

Table 1. Verification summary for final trim-tab roll-control design.

Overall, the completed project produced a feasible and well-defined roll-control design, but not a fully flight-qualified system. The strongest results of the project are the final trim-tab architecture, compact mechanical integration, internal component layout, and early actuation proof of concept. The main limitations are the need for improved linkage stiffness, more precise neutral calibration, aerodynamic load testing, and closed-loop control validation.

The final design gives SCRT a strong foundation for future work. Before flight use, the system should be tested under realistic aerodynamic loading to measure trim tab effectiveness, servo torque margin, linkage deflection, and control response. The next design iteration should also replace temporary or flexible linkage solutions with a rigid mechanical connection that reduces wobble and improves repeatability. Once these improvements are made, the system can be integrated with SCRT avionics for closed-loop testing and eventually evaluated in a sub-scale flight environment.

5 LOOKING FORWARD

In future iterations of the design, the SCRT club may design with attitude control in mind rather than as an addition to a pre-existing rocket design. This would allow the team a greater range of attitude control systems to consider from the outset, rather than working within the geometric and mass constraints from a vehicle designed without an active roll control mission.

If SCRT were to improve upon the current design, the first recommended change would be the addition of a secondary trim connector. The current single-contact interface, while functional, introduces risk of misalignment, loosening under repeated loading, or tab damage over the course of a flight campaign. A second point of contact would better constrain the tab relative to the fin and distribute mechanical loads more evenly. The Roll Control team had reasonable success with the current connector approach (aluminum wedged into a 3D-printed filament housing) but this solution is best understood as a proof-of-concept. A future iteration should investigate more durable interface materials and a retention mechanism capable of withstanding the acting forces during flight.

On the electronics side, a slimmer PCB or proto-board form factor would reduce internal volume consumption and improve packaging flexibility within the airframe. Future teams should also consider integrating the avionics more tightly with the servo actuation system, potentially moving toward a purpose-designed PCB rather than a general-purpose proto-board, which would reduce wiring complexity and improve reliability under vibration. If roll rate sensing is incorporated into future designs, attention should be paid to sensor placement and isolation from structural vibration sources, as signal noise was a notable challenge in ground testing.

More broadly, future teams would benefit from establishing aerodynamic and structural simulation workflows earlier in the design process. Validating trim tab sizing and hinge moment estimates through CFD or flight simulation prior to fabrication would reduce the number of design-test-revise cycles and give the team higher confidence heading into competition.

6 CONCLUSIONS

The Roll Control Capstone team set out to design, prototype, and evaluate an active roll stabilization system for the SCRT competition rocket. Through a structured engineering design process involving concept generation, trade studies, sponsor-reviews, iterative CAD development, and prototype manufacturing, the team selected servo-actuated trim tabs as the most effective and practical solution for providing aerodynamic roll control within the projects geometric, operation, tie, and monetary constraints of the SCRT club.

The completed project resulted in a functional roll-control prototype with integrated CAD architecture capable of demonstrating the feasibility of a trim tab roll control system. The team successfully integrated servos, trim tabs, electronics, and housing/mounting systems into the SCRT rocket fincan and surrounding airframe while maintaining compatibility with existing rocket geometry and systems, minimizing impact to mass and the aerodynamic profile. Testing has verified servo-actuation, trim tab motion creating deflections based on fincan orientation, neutral positioning fail instances, and system integration. The project has not been able to undergo flight testing, but has established a practical and manufacturable foundation that can be expanded upon by the SCRT club, who will be using canards for their future rockets. Though the SCRT plans on integrating canards in future rocket iterations, the developed trim-tab prototype will provide transferable design knowledge and supporting infrastructure to their future systems. The club will benefit from system integration methods, control code architecture, servo-actuation concepts, and housing components designed for this project. As a result, the work completed by the Roll Control Capstone team will serve as both a proof of concept for active aerodynamic control and provide technical documentation and design for future SCRT aerodynamic control system development.

Throughout development, the team used iterative design methods and sponsor-guided downselection to evaluate alternatives and address manufacturing and design issues. Multiple design iterations provided improvements to the manufacturability, and integration of the system into existing geometries/architecture.

Because the trim tab subsystem is intended for use within a high-power rocket that is launched to compete in the annual International Rocket Engineering Competition (IREC), there are important standards and requirements for safety regarding a roll-control system. The most prominent requirement being that during failure of the rocket during flight, all systems must revert to a neutral state. This system has been designed and tested to ensure that during a system failure, the trim tab system will return and remain in a neutral state.

Although the final system is not able to be flight-tested, the project was successful in producing a feasible and well-documented roll-control system that addresses the sponsor’s need for an active roll-control system and has provided SCRT with a strong technical foundation for future testing and integration to a canards based system for future rockets.

7 REFERENCES

[Include here all references cited, following the reference style described in the Style Guide. There should only be one Reference list in this report, so all individual section or subsection reference lists must be compiled here with the main report references. If you wish to include a bibliography, which lists not only references cited but other relevant literature, include it as an Appendix.]

[1] SCRT Roll Control Capstone Team, “Sponsor Meeting Notes: SCRT Roll Control Requirements and Constraints,” Meeting Notes, Oregon State University, Corvallis, OR, Jan. 22, 2026.

[2] SCRT Roll Control Team, SCRT Rocket Fin Can Assembly CAD Model, Unpublished CAD Model, 2026

[3] Posta, Alexandra, et al. “Improved Attitude Stabilisation System Augmented Sounding Rocket Design, Integration, Verification & Launch.” IFAC-PapersOnLine, vol. 58, no. 16, 5 Sept. 2024, pp. 175–180, www.sciencedirect.com/science/article/pii/S2405896324012515, https://doi.org/10.1016/j.ifacol.2024.08.482.

[4] Youds, Theodore, et al. DEVELOPMENT of a CANARD CONTROLLED ACTIVE STABILITY SYSTEM for a SOUNDING ROCKET.

[5] IREC, “International Rocket Engineering Competition Rules & Requirements Document,” Mar. 04, 2025. https://www.soundingrocket.org/uploads/9/0/6/4/9064598/irec_rules_and_requirements_document_v_1.6_.pdf (accessed May 31, 2026).

8 APPENDICES

Appendices contain supporting technical documentation and information referenced throughout the report and provide additional detail regarding the design, manufacturing, and evaluation of the Roll Control system

8.1 Appendix A: Final Cad Assembly Views

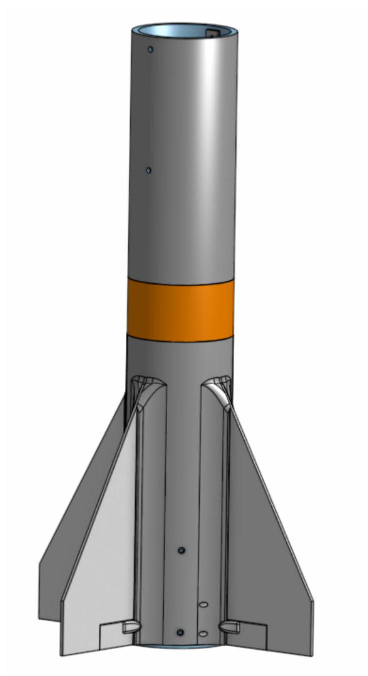

Figure 2. Final CAD Assembly

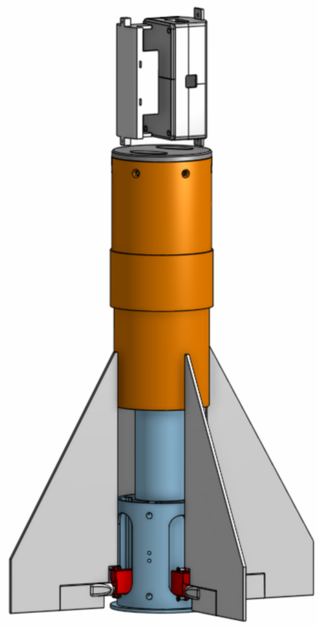

Figure 3. Internal Components of Final CAD Assembly

8.2 Appendix B: Bill of Materials (BOM)

Category	Name	Link	Description	Qty	Price	Notes	Max Rating
Sensor + MCU	Teensy 4.1	https://www.sparkfun.com/teensy-4-1.html	MCU	1	$31.40	500MHz	
Sensor + MCU	BNO 055	https://www.adafruit.com/product/2472?srsltid=AfmBOopnU4S5KTyt03sFgJJ9JqI_LDzZLb13DwnX9TI_5h72Vbb-9_8h	IMU	1	$35.00	handles sensor fusion internally	16g, 2000deg/sec
Sensor + MCU	BMP 280	https://www.digikey.com/en/products/detail/adafruit-industries-llc/2651/5604371	Barometer	1	$9.95		9000m max alt
Power	UBEC	https://www.hobbywingdirect.com/products/ubec-10a-hv-3s-14s?srsltid=AfmBOooR4LccpDV-TUhU1ioaeb4tlV0m_rGZJnKIx4a1gmH9A2DKhD50	ubec	1	$39	super flexible part could be used on future products	10A continuous 25A peak
Power	3S Lipo	https://genstattu.com/gens-ace-2200mah-3s-11-1v-25c-g-tech-lipo-battery-pack-with-ec3-deans-and-xt60-adapter-for-rc-plane/	3s 2200mAh 25C	1	$27	would run the servos at max current for 13 mins	Max current of 55A
Passives	2.2mF Cap	https://www.digikey.com/en/products/detail/nichicon/UVR1V222MHD/588827	Electrolytic Cap	8	$9	for servo power decoupling	

Misc.	Protoboard	https://eecs.engineering.oregonstate.edu/education/tekbotSuite/tekbot/publicInventoryPart.php?stocknumber=P1421783775	3.5x2 inches			Evan can buy from tekbots	
Misc.	Wire		22 AWG			Evan can buy from tekbots	
Misc.	Caps		1uF X7R			Evan can buy from tekbots	
Misc.	Connectors		screw terminals			Evan can buy from tekbots	
Controller	MKS-HV6150	https://www.mksservosusa.com/product.php?productid=284&cat=6&page=1	Servo	4	$285.96	5mm Spline	Stall torque of 6.3Kg/cm
		Total Price - $437.31					

8.3 *Appendix C: House of Quality (HOQ)*

Customer Requirements (CRs) to Engineering Specifications (ESs)				
CR#	**CR description using complete sentences**	**Weight (100 total)**	**Matching Engineering Specification**	**Targets with Tolerances**
1	**The system shall reduce uncontrollable roll to a stable, controllable, low roll rate during flight**	**15**	**The final trim-tab control approach shall demonstrate predicted roll-rate reduction compared to an uncontrolled baseline.**	**Predicted average roll rate reduced by a factor of 10 from uncontrolled baseline. Minimum acceptable reduction: factor of 2.**
2	**The system shall provide sufficient roll-control authority to counter expected aerodynamic rolling moments.**	**12**	**The roll-control system shall provide a controllable rolling moment greater than the estimated worst-case aerodynamic rolling moment.**	**Available control moment shall be at least 50% greater than the estimated worst-case rolling moment, with ±10% acceptable uncertainty.**
3	**The system shall comply with competition safety requirements**	**15**	**The trim tab shall return to or remain near neutral during inactive, startup, or fail-safe conditions.**	**Trim tab neutral offset shall remain within ±2 degrees of neutral.**
4	**The system shall respond**	**10**	**The control system**	**Sensor-to-servo response time shall**

	quickly enough to correct roll disturbances during operation.		**shall respond within 100ms of a roll rate disturbance**	**be ≤100 ms, with maximum acceptable delay of 110 ms.**
5	**The system shall be lightweight enough to avoid significant rocket performance penalties.**	**8**	**The total installed roll control system mass (hardware, wiring, and mounts) will remain no greater than 2.5% of liftoff mass**	**Total subsystem mass shall be <2.5% of rocket liftoff mass**
6	**The system shall fit within the available rocket volume and not interfere with the motor casing, fincan, or fore airframe.**	**12**	**The roll-control subsystem shall fit within the final rocket packaging envelope without interference.**	**System shall use <80% of available packaging volume and have no physical interference with the motor casing, thrust plate, fincan, fore airframe, or moving trim tab.**
7	**The system shall be robust enough to withstand handling, repeated actuation, and expected launch-related mechanical loads.**	**10**	**The clamp, servo-clamp coupler, and trim tab shall remain secure during repeated operation and handling checks.**	**Mechanism shall complete 25 full-range actuation cycles with no clamp slip, coupler loosening, binding, cracking, or visible damage.**
8	**The electronics and power components shall remain secure and functional.**	**8**	**The battery housing and protoboard housing shall securely retain and protect components in the fore airframe.**	**Battery and protoboard shall show no movement, electrical disconnect, shorting, or visible damage after handling/shake test.**
9	**The control system shall command the trim tab in the correct direction and avoid excessive tab motion.**	**6**	**The MCU, sensor, and software shall command the servo in the correct direction and limit trim-tab deflection.**	**Correct servo direction in 100% of bench test cases; maximum commanded tab angle limited to ±15 degrees.**
10	**The final design shall be feasible to manufacture, assemble, inspect, and present as a final subsystem prototype/display.**	**4**	**The final roll-control subsystem shall be represented by complete CAD, physical prototype/display evidence, and a cost review.**	**Final CAD, physical display/prototype, and BOM shall be completed before the final design meeting; prototype cost shall not exceed approved budget.**
	Sum (should be 100)	**100**		

8.4 Appendix D: Trim Tabs and Spincan Trade Study

Trim Tabs	**Active:** servo-moved tabs on fins create aerodynamic torque; strong authority at high speed	**low-medium** (tabs are light, servo size depends on max dynamic pressure on hinge loads)	**Medium** (Servos + linkages + controller tuning)	Best in boost/max dynamic pressure, weaker at low speed/high altitude	**Medium** (The servo and servo hinge are failure points)	**Low-Moderate**
Fincan / Spin-can	**Mostly passive:** reduces roll from aero coupling by letting fin module rotate freely; not precise steering	**Medium** (bearing + stronger tail module)	**High** (mechanical rotating assembly and retentions)	Best if roll is driven by aero coupling	**Medium-High** (reliable if designed well but bearing is the critical point)	**Moderate**

8.5 Appendix E: Spinnerons and Cold-Gas Thrusters Trade Study

System	Roll Control Authority	Mass	Complexity	Applicability	Reliability	Cost
Spinnerons	Passive system for limiting rolling moment, lacking precision though	Average mass around 0.1 kg	Passive roll control, designing a fin with a hinging trailing edge, main challenge is finding best geometry of fin and tab for induced roll control	Lower altitudes and lower velocities do not negatively affect roll control capabilities	Very few failure points, only real concerns are the tabs hinge on the fin and the strength of connection and strength of tab material itself	Likely under one hundred dollars, post design not much iteration should be required and prototyping along with final product should be easily manufacturable
Cold-Gas Thrusters	Low thrust with high precision, with easily controlled on/off release of gas	Average mass around 1-1.5 kg	Requires 3 components: Cold-Gas storage, valve to control release, thruster, and plumbing to release gas from tank to thruster	Lower altitudes require bigger thrusters to utilize roll control	With multiple parts with failure points, and the low altitude aerodynamic effects on the cold-gas thrusters, a lot of iteration and tuning would likely be required	Likely over a few hundred dollars, requiring multiple mechanical components for construction such as valves, nozzles, pressure

						tanks, etc.

8.6 Appendix F: Canards and Differential Fins Trade Study

System	Roll Control Authority	Mass	Complexity	Applicability	Reliability	Cost
Canards	Active Control:	Low: canards are small, the rest of the weight would be in the servos	medium: requires active control system with sensor and motor drivers	Low: Could cause aerodynamic issues	The control system and servos are points of failure	Medium: Cost of physical canards is low, but the cost of servos is high
Differential Fins	Active Control:	medium: fins could be large depending on design	medium: requires active control system with sensor and motor drivers	Low: could interfere with existing systems on the rocket	The control system and servos are points of failure	Medium: Cost of physical fins is medium, but the cost of servos is high

8.7 Appendix G: Rollerons and Thrust Vectoring Trade Study

	Validity	Cons	Benefits	Specifics	Weight Changes
Rollerons	Potential, but unlikely	No ECE element, Requires velocity before activation/active roll control, which means either the portion of the initial flight will be un-controlled roll, or another component will need to be created to spin the flywheels before launch	No change to internal systems, Minimal change to CG & profile. No ECE elements	~0.5 lbs x4	2lbs total
Thrust Vectoring	Unlikely	Highly complex, requires space and a large change in mass	Could be used to control overall attitude,	--	--